Robotics, Artificial Intelligence and Embedded Systems

Cognitive Systems

Neurons and the Brain I

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Topics in this Chapter

The basic building blocks…
… of the human brain

- What is the anatomical organization of the human brain?
- How is cognition reflected in the structure of the brain?
- How to model and simulate the brain?

The Human Brain

Cognition – The State of the Art

≈ 13 Watts

Modular

Adaptive

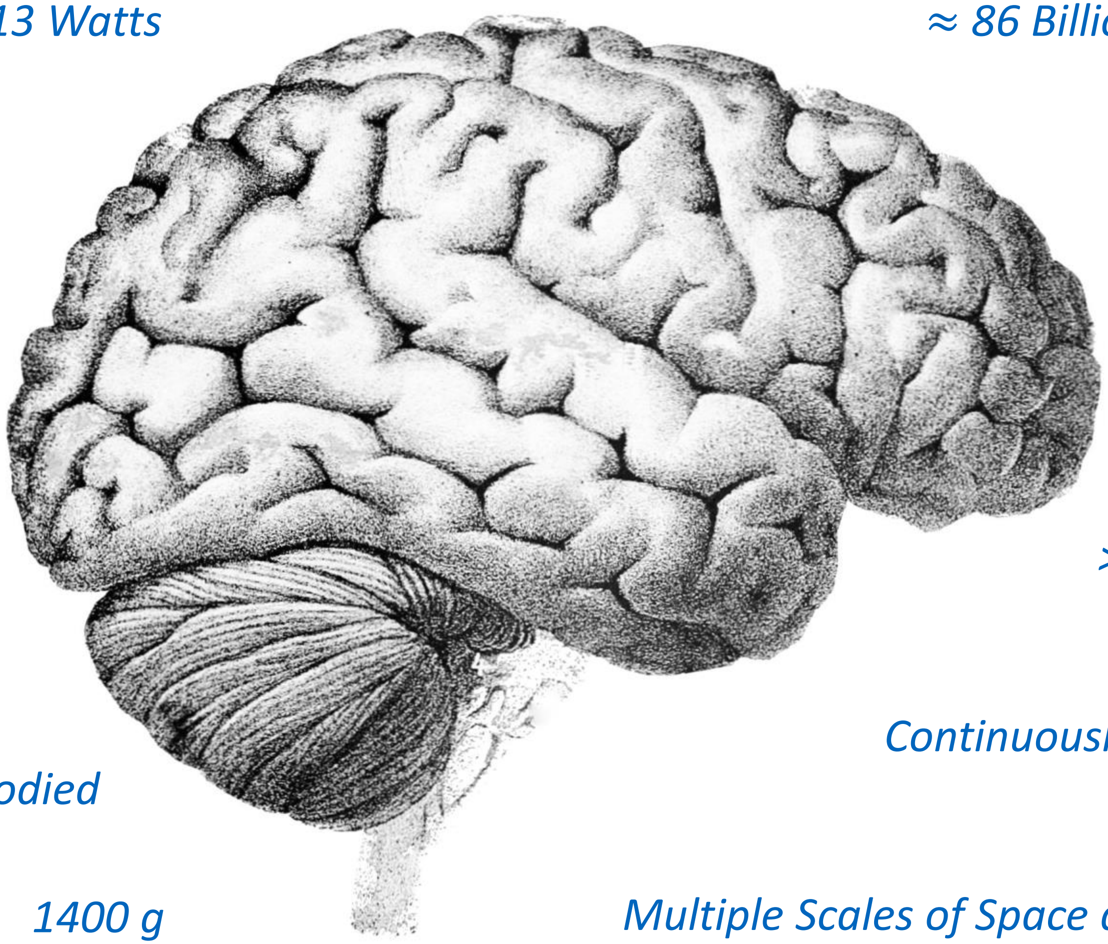
Responsive

Distributed

Robust

Embodied

1400 g



≈ 86 Billion Neurons

180 Brain Regions per Hemisphere

150 Trillion Neocortical Synapses

> 750.000 km of Nerve Fibers

Continuously Changing Structure

Multiple Scales of Space and Time

The Vertebrate Nervous System

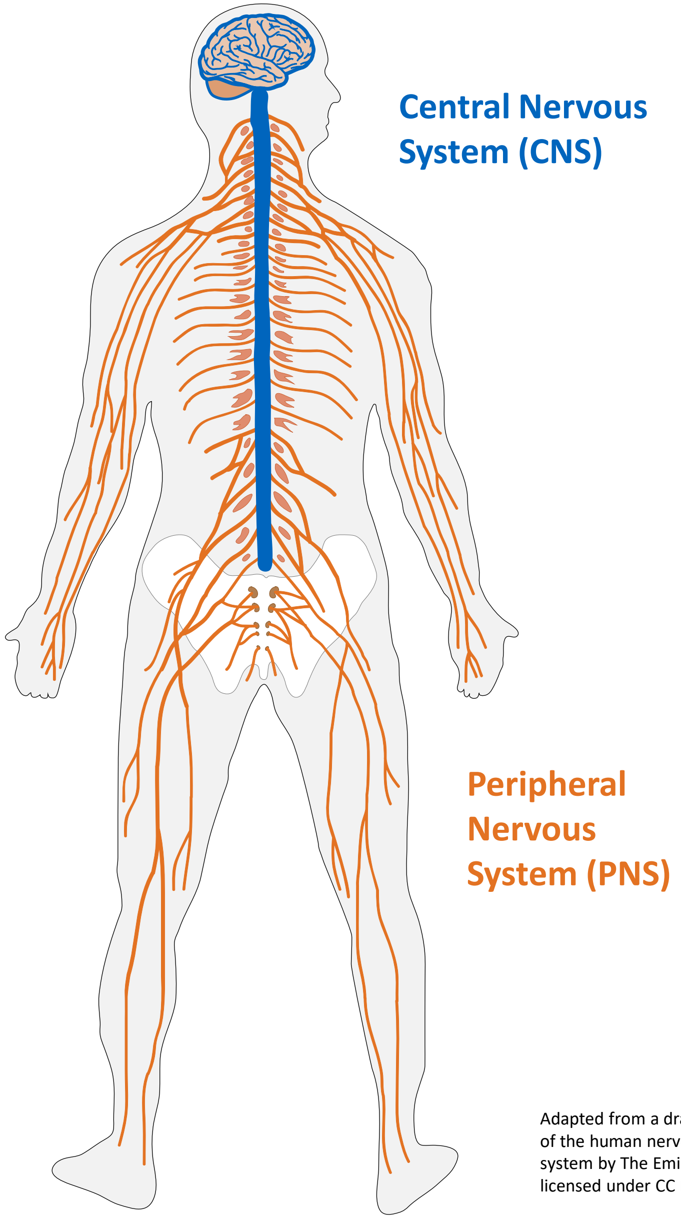
The nervous system is an “organized group of cells specialized for the conduction of electrochemical stimuli from sensory receptors through a network to the site at which a response occurs.” In vertebrates, the nervous system is comprised of two subsystems:

Central Nervous System (CNS) (Zentralnervensystem)

The CNS is the central information processing system formed by the brain and the spinal cord. It collects and distributes data throughout the body.

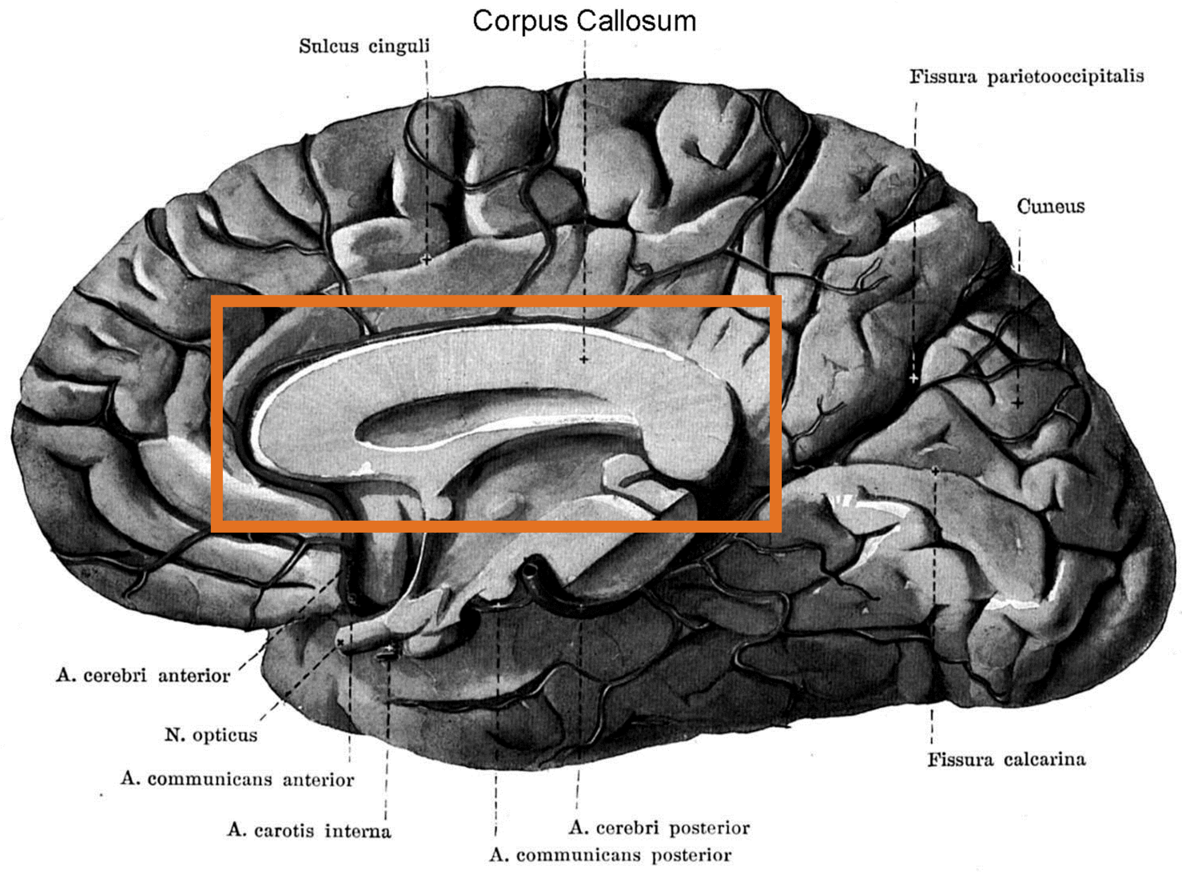
Peripheral Nervous System (PNS) (Peripheres Nervensystem)

The peripheral nervous system transmits signals between sensory organs, muscles and internal organs and the CNS.



Adapted from a drawing of the human nervous system by The Emirr licensed under CC BY 3.0

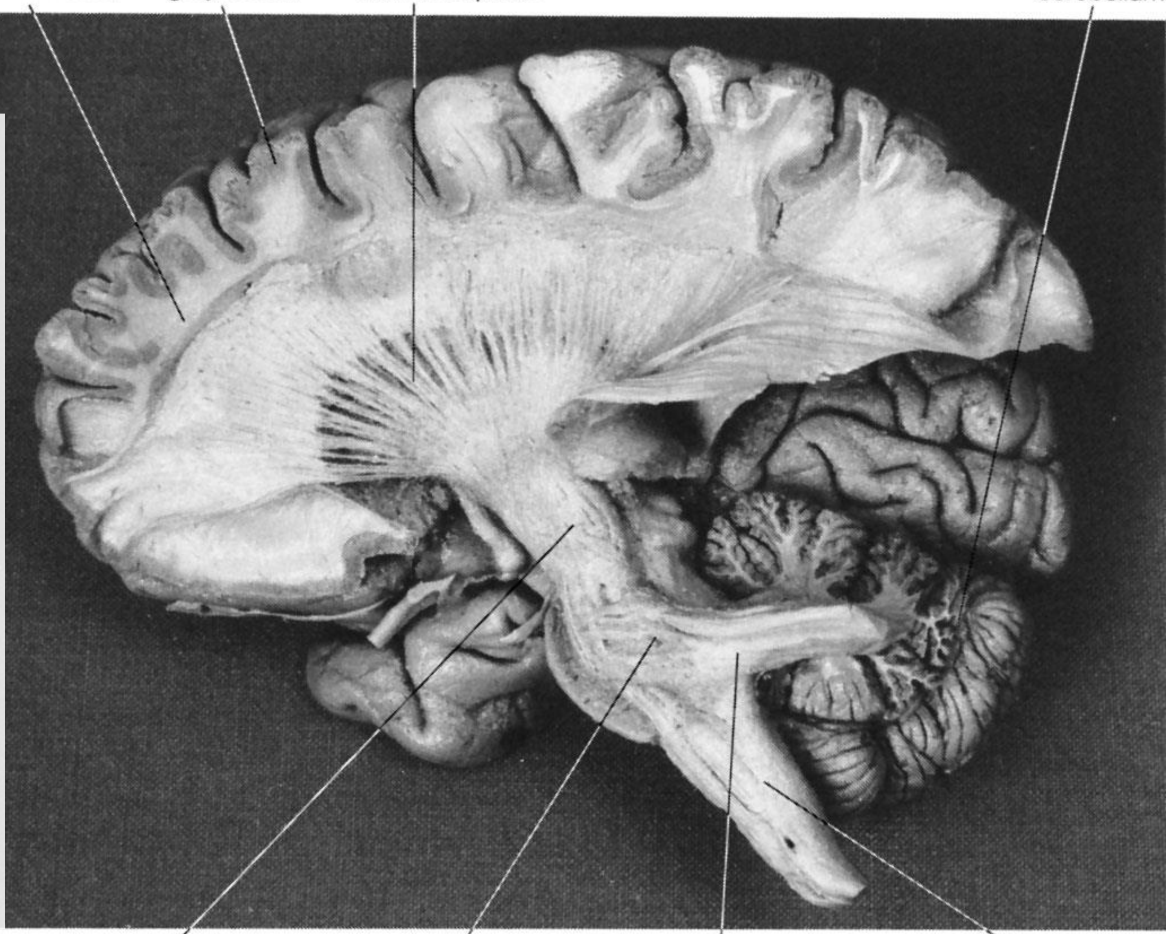
The Anatomy of the Human CNS: Cerebrum



The cerebrum (Großhirn) is the largest and uppermost portion of the brain and accounts for sensory integration, voluntary motion and higher-level cognitive functions.

- It is split into two hemispheres that are connected by the corpus callosum (Balken).

The Anatomy of the Human CNS: Cerebrum



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- It is split into two hemispheres that are connected by the corpus callosum (Balken).
- The cerebral cortex (Großhirnrinde), also called the gray matter (graue Substanz), is the folded outer layer of the cerebrum that is mainly comprised of cell bodies.
- The inner part of the cerebrum, the white matter (weiße Substanz), is a core of nerve fibers that connect the cortical regions.

The Anatomy of the Human CNS: Cerebrum

The surface of the cerebral cortex is subdivided based on three anatomical features:

Gyri (sg. Gyrus) (Windungen)

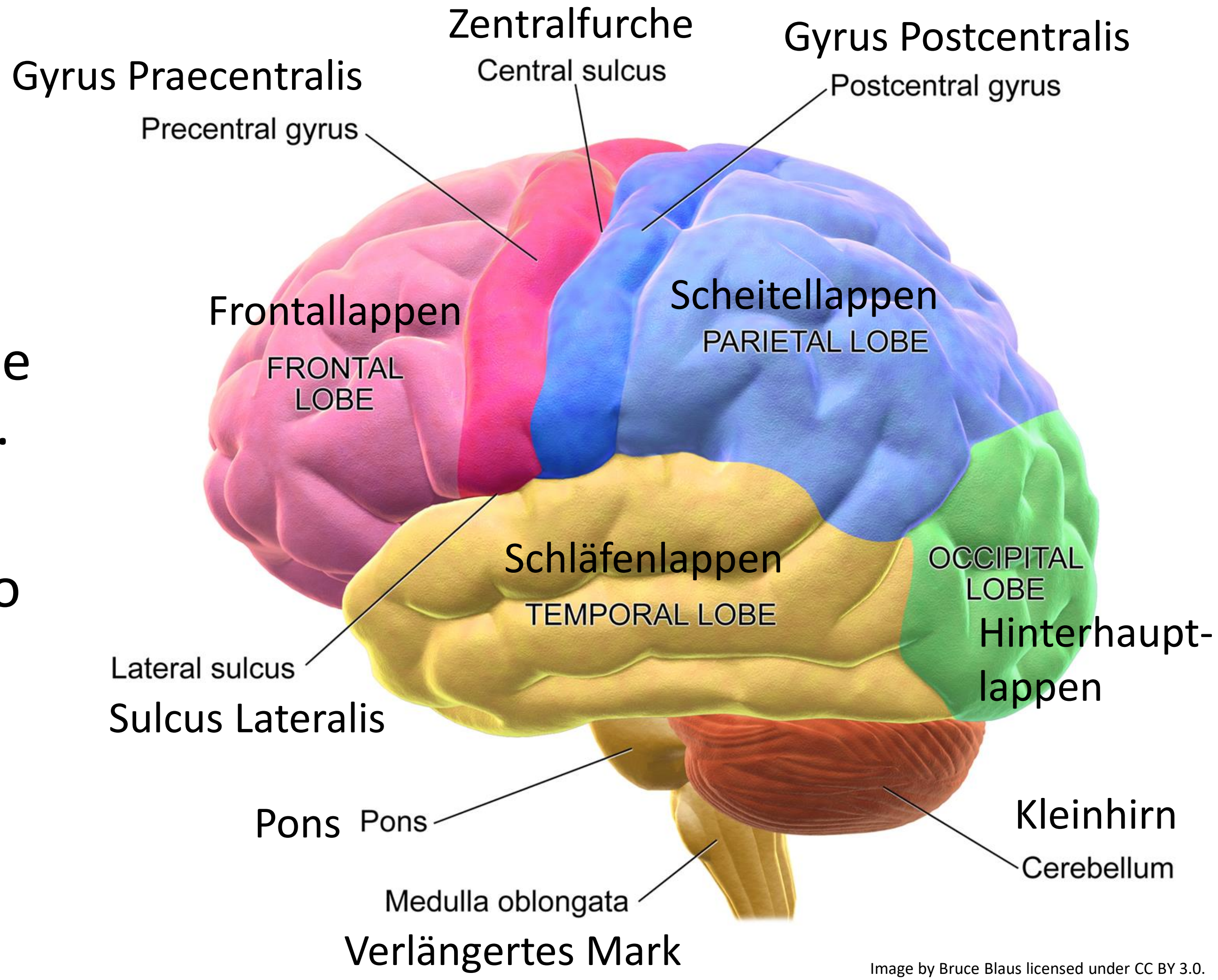
Gyri are the crests (Kämme) formed by the convoluted surface of the cerebral cortex.

Sulci (sg. Sulcus) (Furchen)

A sulcus is the fissure (Spalt) between two neighboring sulci.

Lobes (Lappen)

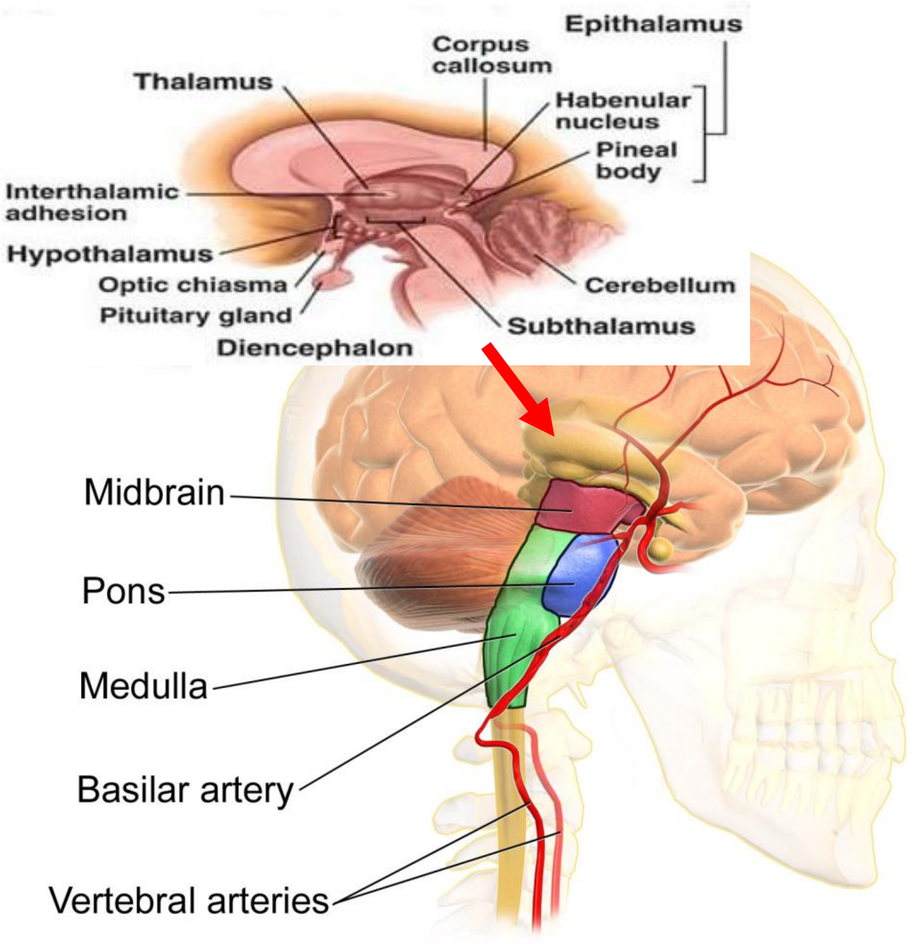
On each hemisphere, two major sulci divide the cerebral surface into a frontal, parietal, temporal and occipital lobe.



Summary: Anatomy of the Cerebrum

From http://blausen.com/en/video/cerebral-landmarks/

The Anatomy of the CNS: Brainstem (Hirnstamm)



Blausen.com staff (2014). "Medical gallery of Blausen Medical 2014". WikiJournal of Medicine 1 (2). Licensed under CC BY 3.0.
https://www.assignmentpoint.com/science/biology/epithalamus.html

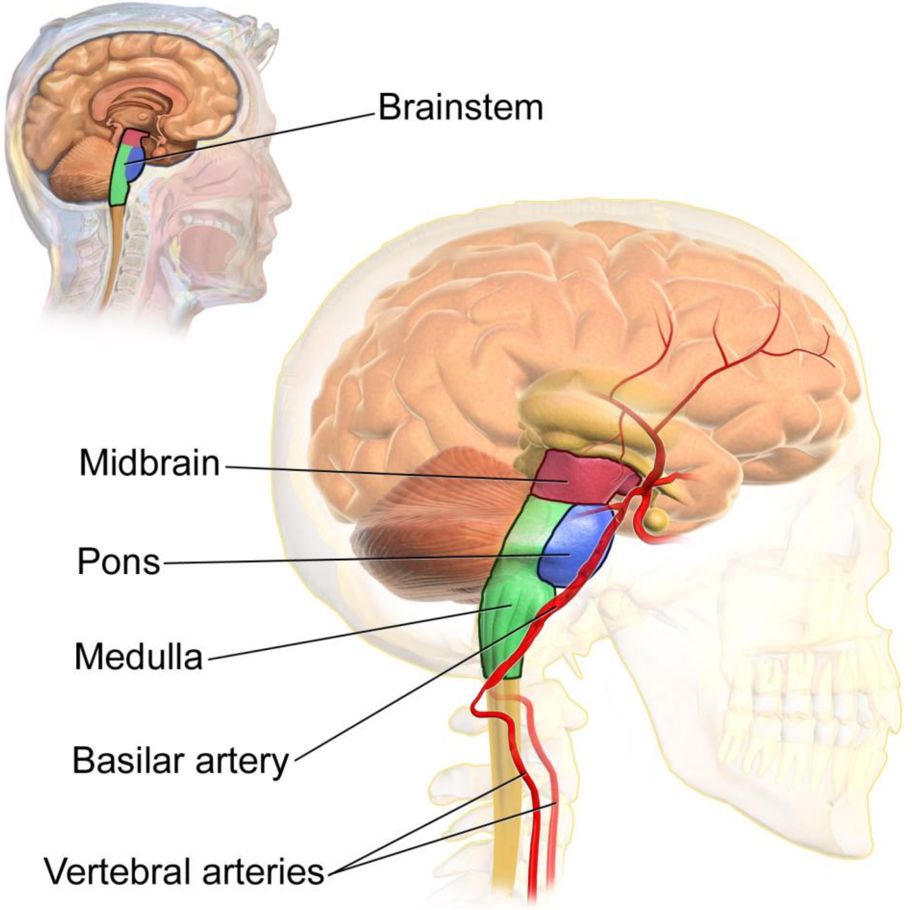
The brainstem connects the cerebrum with the spinal cord. It is divided into four sections:

Diencephalon (Zwischenhirn)

The diencephalon consists of four sub-structures:

- Epithalamus: production of the hormone melatonin (biological clock)
- Thalamus: relay and distribution of sensory and motor signals to the different regions of the cerebral cortex
- Hypothalamus: Control of autonomic functions (temperature regulation, appetite), behavior, and hormone production
- Subthalamus

The Anatomy of the CNS: Brainstem (Hirnstamm)



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Midbrain (Mesencephalon) (Mittelhirn)

The midbrain is involved in the control of eye movements (including visual reflexes for e.g. object tracking), auditory processing, and visual processing.

Pons

The pons is comprised of nerve fibers that connect the medulla oblongata and the cerebral cortex with the cerebellum. It further transmits sensory and motor signals between the brain and the facial region.

Medulla Oblongata (Verlängertes Mark)

The medulla oblongata connects the brain with the spinal cord.

There are different schemata for denominating the individual components of the brain. For example, pons, medulla oblongata and the cerebellum (Kleinhirn) form the hindbrain (Rautenhirn).

Overview: Brain Stem

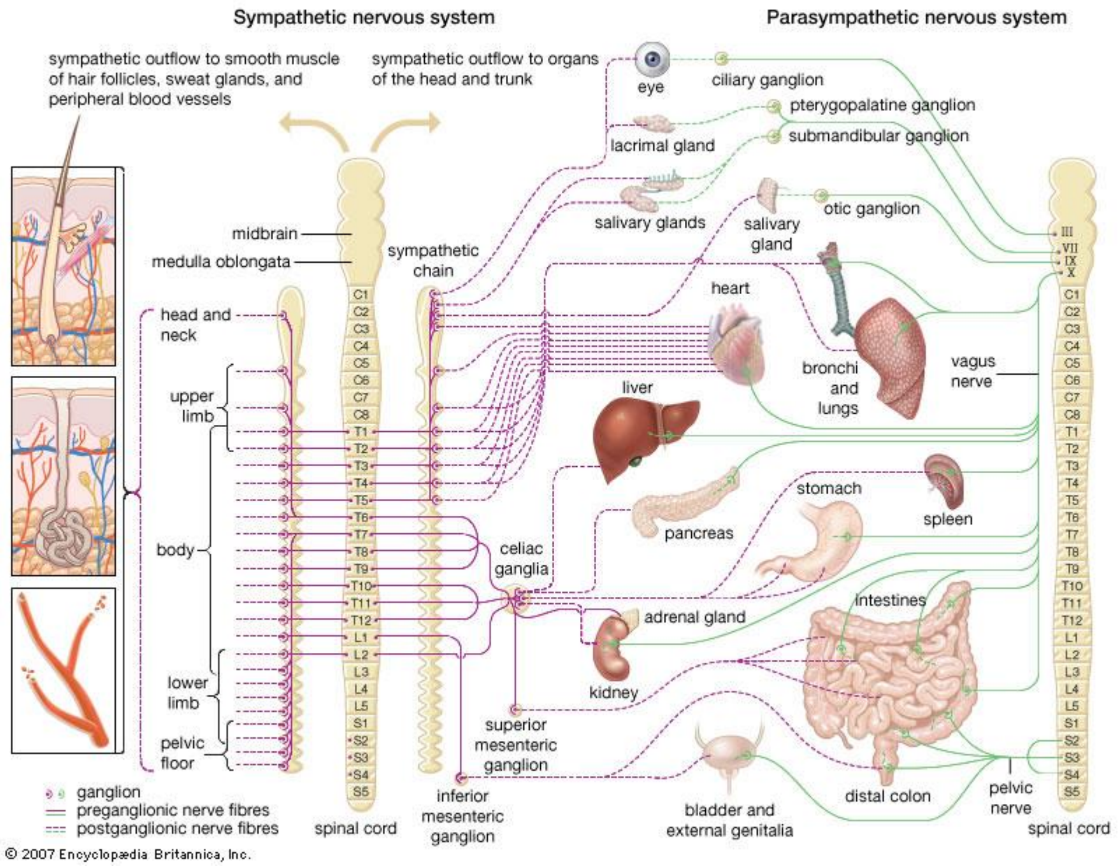
From http://blausen.com/en/video/brainstem/

The Human Peripheral Nervous System

Autonomic Nervous System (Vegetatives Nervensystem)

The autonomic nervous system is responsible for homeostasis (self-regulation) and operates largely unconsciously. It includes two antagonist subsystems:

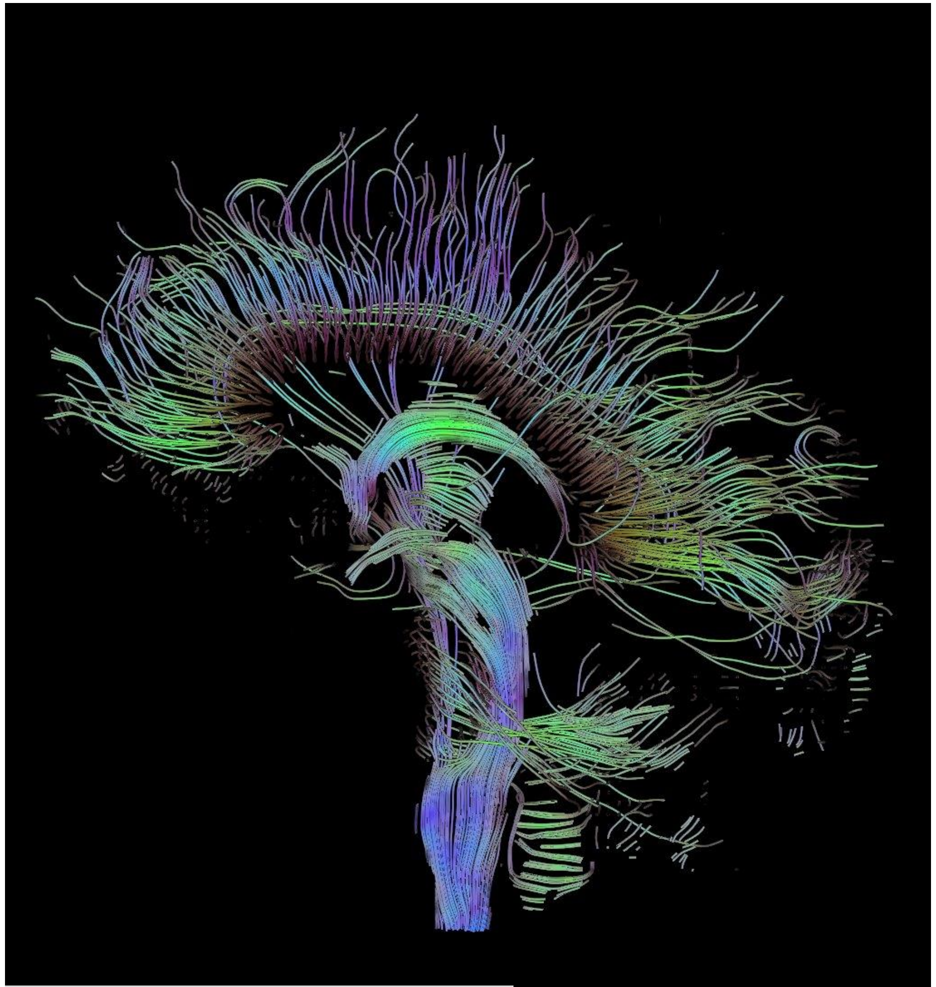
- Sympathetic nervous system (Sympathikus): connection of the internal organs to the brain; preparation of the organism for stress (e.g. increase of heart rate and blood flow to the muscles, sweating due to heat)
- Parasympathetic nervous system (Parasympathikus): mainly cranial nerves (Hirnnerven) and lumbar (Lenden-) spinal nerves (Rückenmarksnerven); sets the body to resting state and increases digestive functions



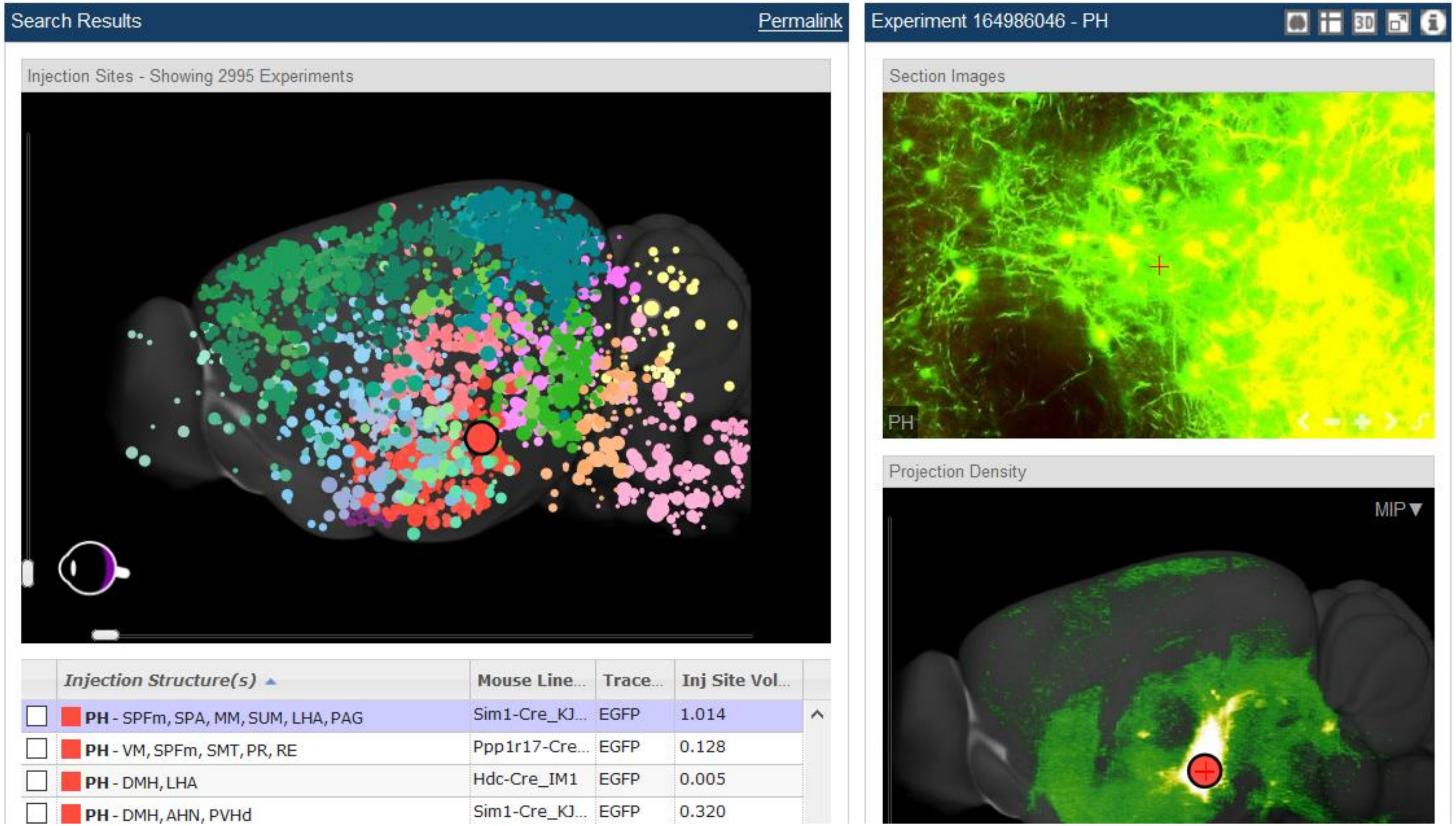
From Encyclopedia Britannica.

Brain Networks

- The brain is a highly complex network of interconnected cells and modules
- Connectivity is both static (anatomical connectivity) and dynamic (functional connectivity, dependent on the current cognitive task)
- Brain networks can be described as graphs and analyzed with graph theory
- The network topology can be analyzed using standard graph-theoretic measures (node degrees, clustering coefficients, hubs, path lengths)
- Brain networks are thought to be related to information integration (integrated information theory)



Demo: Connectome Data in the Allen Mouse Brain Atlas



http://connectivity.brain-map.org/projection

Cognition without Brains

Simpler species such as jellyfish have a diffuse nervous system that is distributed throughout the body.

Sponges have no neurons at all.

The endocrine system (Hormonsystem) that is also present in humans is a regulatory systems that works in parallel to the nervous system by secreting hormones.

Some Myths about the Human Brain

Only ten percent of the brain are used

Every part of the brain is used but not all parts are used at once (which would result in epilepsy).

The brain does not produce new neurons in adults

In certain brain areas, new neurons are produced throughout life.

The size of the brain is related to intelligence

The brains of elephants and whales are much larger than those of humans.

Brain Sizes

Roth, G., & Dicke, U. (2005). Evolution of the brain and intelligence. *Trends in cognitive sciences*, *9*(5), 250-257.

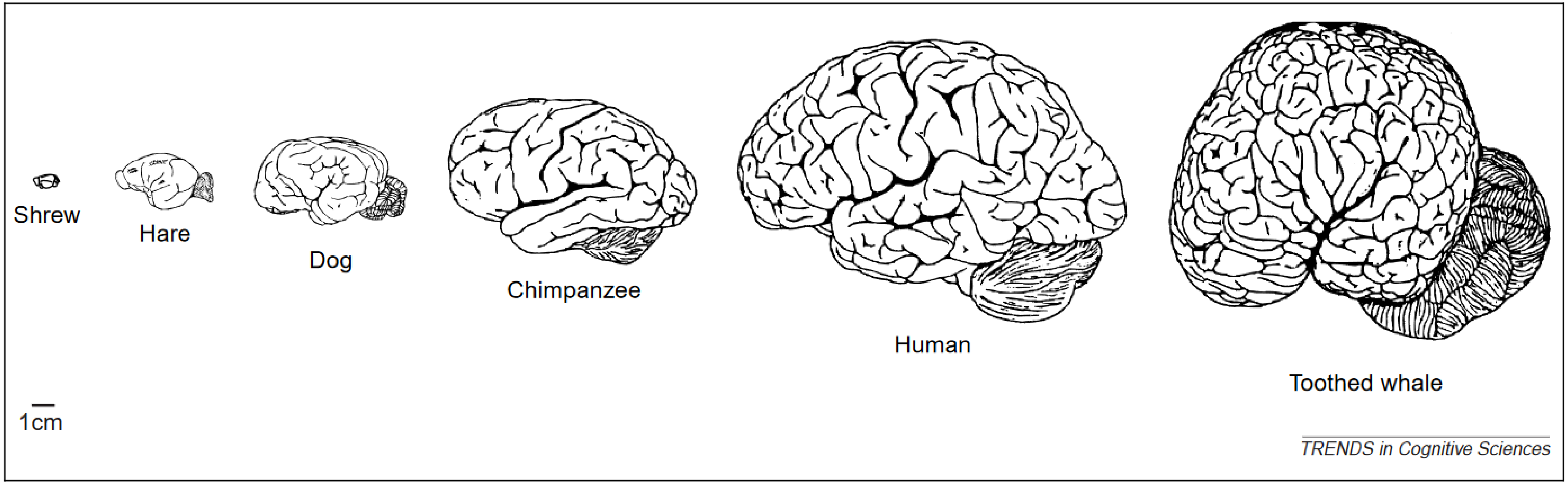


Figure 1. A series of mammalian brains. Humans do not have the largest brain in absolute terms and are exceeded in size by many cetaceans (whales, dolphins, porpoises) and the elephants. They also do not have the most convoluted cortex. With a few exceptions, convolution of the cortex increases in proportion to cortical size.

The Encephalization Quotient

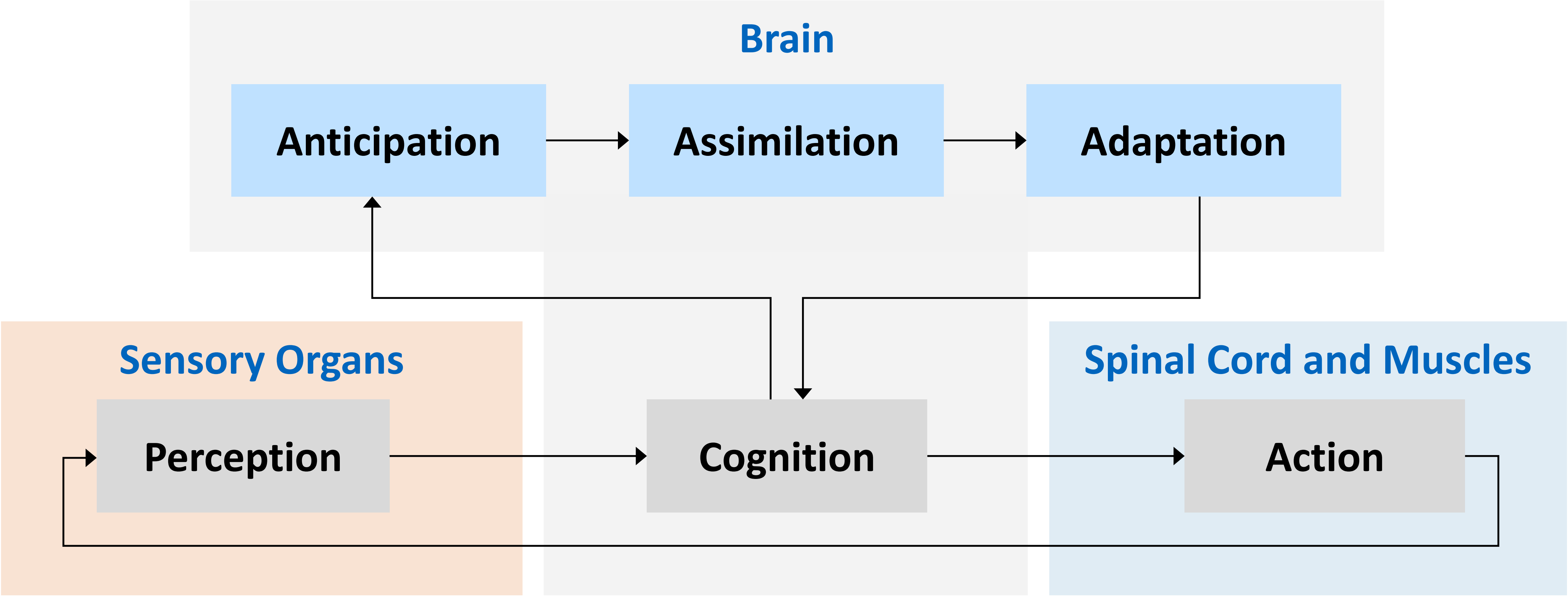
Roth, G., & Dicke, U. (2005). Evolution of the brain and intelligence. *Trends in cognitive sciences*, *9*(5), 250-257.

Table 1. Brain weight, encephalization quotient and number of cortical neurons in selected mammals

Animal taxa	Brain weight (in g)[a]	Encephalization quotient[b,c]	Number of cortical neurons (in millions)[d]
Whales	2600–9000	1.8	
False killer whale	3650		10 500
African elephant	4200	1.3	11 000
Man	1250–1450[e]	7.4–7.8	11 500
Bottlenose dolphin	1350	5.3	5800
Walrus	1130	1.2	
Camel	762	1.2	
Ox	490	0.5	
Horse	510	0.9	1200
Gorilla	430[e]–570	1.5–1.8	4300
Chimpanzee	330–430[e]	2.2–2.5	6200
Lion	260	0.6	
Sheep	140	0.8	
Old world monkeys	41–122	1.7–2.7	
Rhesus monkey	88	2.1	480
Gibbon	88–105	1.9–2.7	
Capuchin monkeys	26–80	2.4–4.8	
White-fronted capuchin	57	4.8	610
Dog	64	1.2	160
Fox	53	1.6	
Cat	25	1.0	300
Squirrel monkey	23	2.3	480
Rabbit	11	0.4	
Marmoset	7	1.7	
Opossum	7.6	0.2	27
Squirrel	7	1.1	
Hedgehog	3.3	0.3	24
Rat	2	0.4	15
Mouse	0.3	0.5	4

[a]Data from [13,17,73].
[b]Indicates the deviation of the brain size of a species from brain size expected on the basis of a 'standard' species of the same taxon, in this case of the cat.
[c]Data after [13,73].
[d]Calculated using data from [17].
[e]Basis for calculation of neuron number.

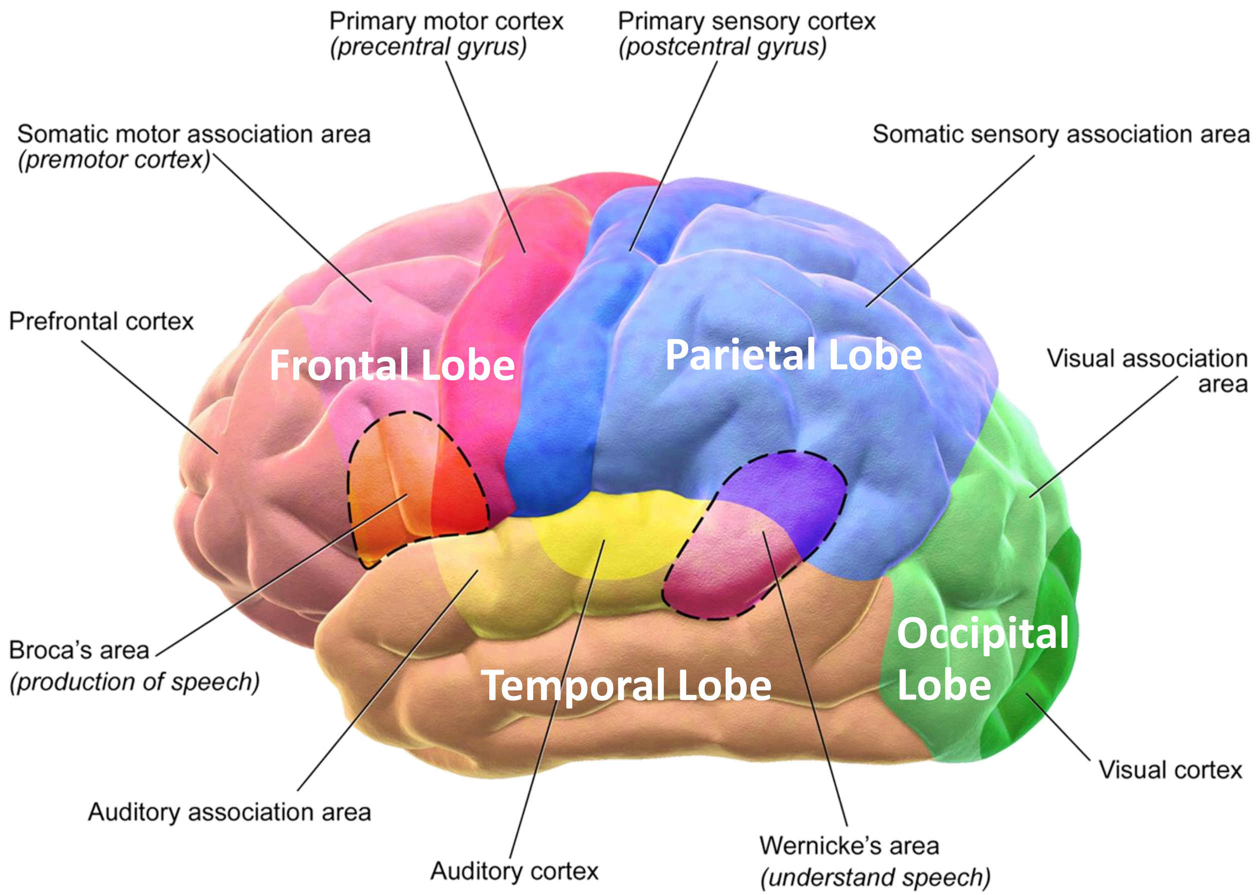
PCA – A “Mapping” to the Nervous System



Cognition: The Cerebral Cortex

The cerebral cortex is the center of higher cognitive functions:

- The mapping between cognitive functions and brain regions (lobes) can be determined with functional brain imaging
- Frontal lobe (Frontallappen): short-term memory, action planning, movement control
- Parietal lobe (Scheitellappen): somatic sensation, body image
- Occipital lobe (Hinterhauptlappen): vision
- Temporal lobe (Schläfenlappen): hearing, learning, memory, emotion



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The Human Neocortex

The neocortex is part of the cerebral cortex and considered to be the center of higher cognitive functions and intelligence. Most parts are organized in six different layers:

Layer I: Molecular Layer (Molekularschicht)

Dendrites of cells located in deeper layers and axons that travel through this layer in other areas of the cortex

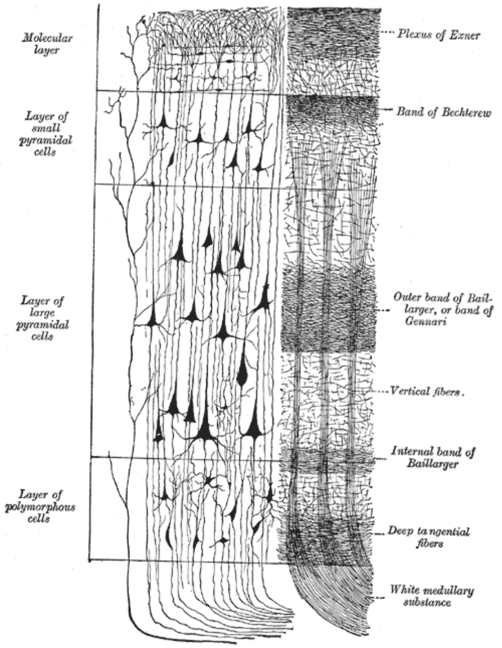
Layer II: External Granular Cell Layer (äußere Körnerzellenschicht)

Small pyramidal-shaped cells. Projections to other cells within the same and to other cortical areas.

Layer III: External Pyramidal Cell Layer (äußere Pyramidenschicht)

Similar to Layer II. However, neurons located deeper in the layer are "typically larger than those located more superficially."

Outside



Inside

The Human Neocortex

Layer IV: Internal Granule Cell Layer (innere Körnerzellenschicht)

Large number of small spherical neurons. Layer IV is the “main recipient of sensory input from the thalamus” and sometimes divided into three sub-layers.

Layer V: Internal Pyramidal Cell Layer (innere Pyramidenschicht)

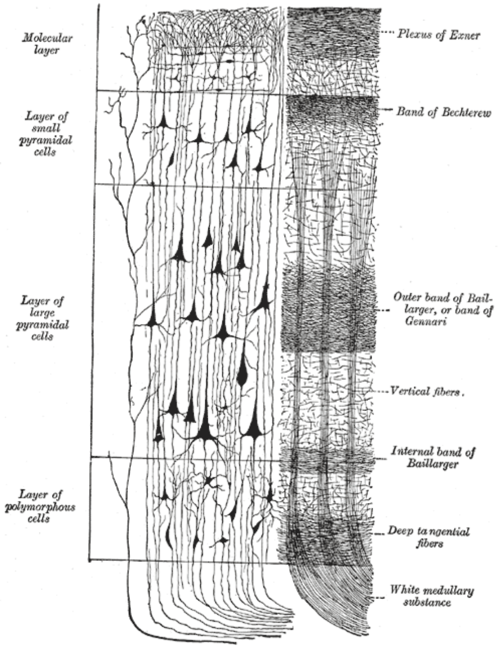
Pyramidal cells that are large than those in Layer III. Neurons in this area project to other cortical areas and subcortical structures and form the major output pathways of the cortex.

Layer VI: Multiform Layer (multiforme Schicht)

Layer with heterogeneous neurons which blends into the white matter.

https://link.springer.com/referenceworkentry/10.1007%2F978-1-4614-7320-6_571-1
https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2527871/
http://www.cs.toronto.edu/~miguel/papers/ps/ecs02.pdf
https://journals.plos.org/ploscompbiol/article?id=10.1371/journal.pcbi.1005045

Outside

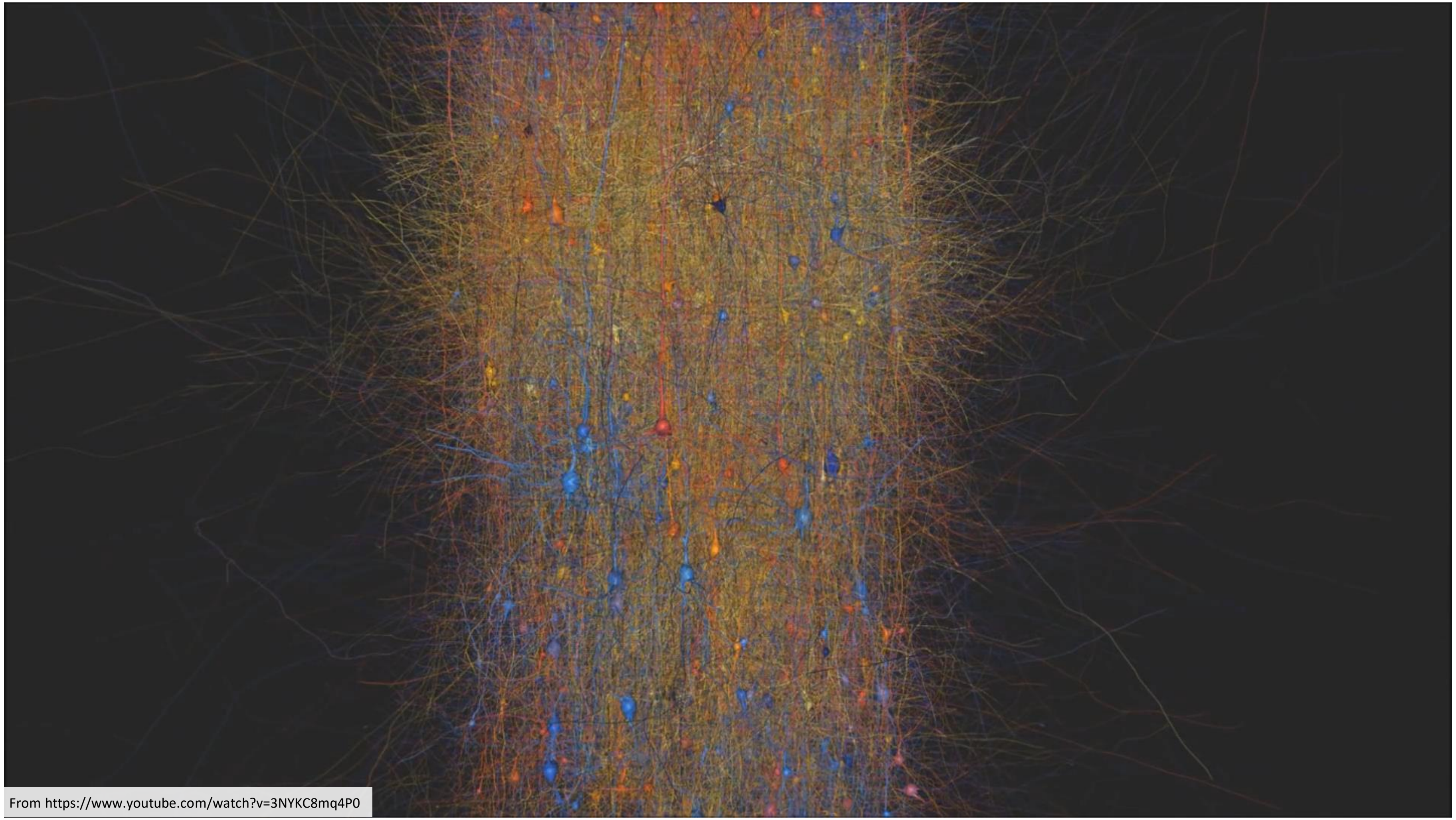


Inside

Cortical Microcircuit

In addition to the laminar structure, the cortex is organized in columns (Kolumne/Spalt/Säule).

The most basic building block is the minicolumn, which consists of 80-100 neurons. Cortical columns or modules are comprised of many minicolumns.



From https://www.youtube.com/watch?v=3NYKC8mq4P0